

The Blind Spot Paradox

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When a self-repairing model destroys the evidence its own monitor needs

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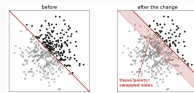
The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

The Blind Spot Paradox

└ Concept drift: the rule changes, the data does not

First, the phenomenon. A model learns a rule: this input goes with that answer. Then the world changes and the rule no longer holds. That is concept drift. Here is the cleanest version of it, and it is what we simulate: two inputs, a straight boundary, a label on each side. The change moves the boundary, and nothing else. Only the shaded band has swapped sides, and a model trained before is now wrong on it.

Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong** on the shaded band.

The Blind Spot Paradox

└ Two defences, built by two separate communities

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- A **model that repairs itself**. A forest of ten small decision trees. When a tree starts failing, it is thrown away and a fresh one takes its place.
- A **watchdog that raises the alarm**. It reads the model's mistakes, adds up the excess, and calls for a human once the pile crosses a threshold.

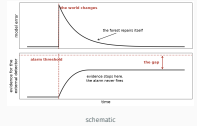
The alarm is the only thing a human ever sees.

Two defences against drift, built by two separate communities. First, a model that repairs itself: a forest of ten small decision trees, and a tree that starts failing is thrown away and replaced. Second, a watchdog: it reads the model's mistakes and calls for a human once the pile of errors crosses a threshold. That alarm is the only thing a human ever sees. Two communities, and nobody had measured their tools against each other.

The Blind Spot Paradox

└ Put them together, and they cancel out

Put the two together, and they cancel out. The boundary moves, the mistakes begin: that is the top curve. The watchdog needs exactly those mistakes. But the forest repairs itself, the mistakes stop, and the pile stops growing before it is high enough. The repair erased the evidence the watchdog was waiting for. The better the model fixes itself, the more surely the alarm stays silent.

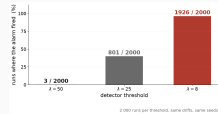


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└ Why not just lower the threshold?

Does the alarm ever fire? That depends on one number: the threshold. We re-ran the benchmark at the paper's three settings, two thousand runs each. At the highest threshold it fired three times out of two thousand. Lower it, and it fires almost every time. So why not lower it? Because a low threshold also fires when nothing has happened, and an alarm that cries wolf gets switched off. The setting quiet enough to be trusted is the one that never sees the drift.

Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

The Blind Spot Paradox

└ Our contribution: the accepted clock is wrong

To study a race you need a stopwatch, and that is where we found something. The forest replaces its trees one at a time. The date the literature uses to say the model has recovered is the moment the very first tree is replaced: one out of ten. Wait until three quarters of the forest has been renewed, and you wait roughly ten times longer. That stopwatch does not time a repair. It times the start of one.

Our contribution: the accepted clock is wrong

called "repaired"
by the literature

86 steps

three quarters of
the forest renewed

1225 steps

0 200 400 600 800 1000 1200 1400
time after the change (steps)

median over 100 runs at $\Delta t = 0.24$, $W = 10$

It marks the first tree out of ten, not a repaired forest.

The Blind Spot Paradox

└─Where we stand

Where we stand

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next**: watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.

Where we stand. The paradox is real and reproducible, and we can say when the evidence falls short. The clock the field relies on is biased, always the same way, so every conclusion drawn from it inherits that bias. What remains is to turn this into advice a practitioner can use. Thank you.